Delayed

Current state: 70°F
Available report: 67°F

Correct about the past.
Late for this decision.

Randomly noisy

Steady state: 68°F
One report: 72°F

Unusual under a stated
noise model, but possible.

Biased

Steady state: 68°F
Typical report: 70°F

A persistent +2°F offset.
Averaging will not fix it.